Quick start: Ames_sub1

Run a complete SWAT+ simulation with the small reference dataset that comes with the source code.

The dataset

`Ames_sub1` is a test dataset of 12 HRUs near Ames, Iowa, with no channels, aquifers or reservoirs. It simulates 1 January 1975 to 31 December 2020 on a daily time step. It is used to test the model and has not been calibrated. The folder is in the SWAT+ source repository at `refdata/Ames_sub1`.

Before running

You need:

- the SWAT+ executable (see Install);
- a copy of the `Ames_sub1` folder, either from a clone of the source repository or downloaded from the link above;
- about 10 GB of free disk space. With the shipped `print.prt` the run writes daily carbon output for every HRU, about 9 GB in 62.0.1. To keep the output small, change the `daily` column of the `hru_cb` and `hru_cb_vars` rows in `print.prt` from `l` to `n` before running.

The shipped folder already contains output files from earlier model versions (for example `basin_wb_aa.txt`, written in 2024). A run with the shipped `print.prt` does not rewrite them, so check the date in the first line of an output file, or delete the old outputs (`*_aa.txt`, `*_yr.txt`, `*_day.txt`, `*.out`) from your copy before running.

Running

From a terminal, change into your copy of the folder and run the executable from there:

```
# Linux and macOS
cd path/to/Ames_sub1
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

```
:: Windows
cd path\to\Ames_sub1
path\to\swatplus-62.0.1-gnu-win_amd64-Rel.exe
```

SWAT+ reads its inputs from the current folder, so always start it from inside the project folder. The run takes a few minutes; on a recent Linux workstation it took 4 minutes 20 seconds, most of it writing the daily carbon output.

Output

While it runs, SWAT+ prints the simulation date to the screen:

```
  Original Simulation            1   14   1994 Yr    20 of    46 Time   12:37:36
```

At the end, the screen and `simulation.out` show:

```
 Execution successfully completed
```

and the same line is written to `success.fin`.

With the shipped `print.prt`, a 62.0.1 run writes:

Files	Content
`hru_wb_day.txt`, `hru_wb_aa.txt`	HRU water balance, daily and average annual
`hru_nb_day.txt`, `hru_nb_aa.txt`, `hru_ncycle_day.txt`, `hru_ncycle_aa.txt`	HRU nutrient balance and nitrogen cycle
`hru_ls_aa.txt`	HRU sediment and nutrient losses, average annual
`hru_pw_day.txt`, `hru_pw_aa.txt`	HRU plant growth and stress
`crop_yld_yr.txt`, `crop_yld_aa.txt`, `basin_crop_yld_yr.txt`, `basin_crop_yld_aa.txt`	crop yields
`hru_carbvars.txt`, `hru_cbn_lyr.txt`, `hru_seq_lyr.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt`, `hru_n_p_pool_stat.txt`, `hru_org_*_vars.txt`, `hru_plc_stat.txt`, `basin_carbon_all.txt`, `hru_begsim_soil_prop.txt`, `hru_endsim_soil_prop.txt`	carbon and soil organic matter
`mgt_out.txt`, `lu_change_out.txt`	management operations performed, land use changes
`simulation.out`, `diagnostics.out`, `checker.out`, `erosion.out`, `files_out.out`, `co2.out`	run log, warnings, input summaries

A few other files appear as well: `paddy_test.csv` (a rice paddy test output written on every run), `wtps_wuses.wal` and an empty file named `null`.

Open `hru_wb_aa.txt` to see the average annual water balance of each HRU. For the first HRU in this run:

precip	surq_gen	latq	perc	et
952.6 mm	34.7 mm	0.1 mm	0.0 mm	932.5 mm

The columns are described in Water balance outputs.

Warnings in diagnostics.out

`diagnostics.out` lists anything the model could not match while reading the inputs. For `Ames_sub1` at 62.0.1 it contains:

```
agrr_lum      AGRR_rot      not found in management.sch
soyb_lum      SOYB_rot      not found in management.sch
oats_lum      OATS_rot      not found in management.sch
wwht_lum      WWHT_rot      not found in management.sch
ames.tem                    file not found (tgage)
```

The first four are land uses in `landuse.lum` that name management schedules missing from `management.sch`; no HRU uses those land uses. The last line means the station's temperature file, `ames.tem` in `weather-sta.cli`, is not in the list in `tmp.cli` (which names `ames.tmp`), so the model generates daily temperatures from the weather generator statistics in `weather-wgn.cli`. The Ames_sub1 walkthrough explains both. Check this file after every first run of a new project.

Troubleshooting

- `Cannot find object.cnt input file`: SWAT+ was started outside the project folder. Change into the folder first.
- `Permission denied` on Linux or macOS: run `chmod +x` on the executable.
- A command or the model stops with no message: check free disk space.
- The run stops early: read `simulation.out`, `diagnostics.out` and `checker.out` for the last message before the stop.

See FAQ and troubleshooting for other errors.

Next

- Project folder anatomy explains what each file in the folder does.

- The Ames_sub1 walkthrough goes through the dataset file by file.
- Your first watershed builds a project for a new area.

Sources

- Dataset: `refdata/Ames_sub1` at 62.0.1 (`object.cnt`: 12 HRUs and no other objects; `time.sim`: 1975 to 2020, daily; `print.prt`: HRU rows only, with `hru_cb` and `hru_cb_vars` daily set to `1`).
- Run: `swatplus-62.0.1-gnu-lin_x86_64-Rel` from the 62.0.1 release, 20 October 2026. Output size and file list are from that run.
- Completion message and `success.fin`: `src/main.f90.in`. `paddy_test.csv` is opened unconditionally in `main.f90.in` as a temporary paddy output.